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Does This Ontological Commitment Make Me Look Fat?

A Socratic Dialogue on Quantum Puzzles

Alice pondered silently for a bit as she refilled her wineglass. “Let me get this straight,” she said at last. “You actually want to talk about the foundations of quantum mechanics?”

“Sure,” replied her father with a mischievous smile. He was a physicist himself, one who had made a successful career as a master of imposing technical calculations in particle physics. Experimentalists who smashed particles together at the Large Hadron Collider would regularly consult him on difficult questions about jets of particles created by decaying top quarks. But when it came to quantum mechanics, he was a user, not a producer. “It’s about time I got a better understanding of my daughter’s own research.”

“Okay,” she answered. In graduate school Alice had initially started down a similar career path as her father, but had gotten sidetracked by a dogged insistence on making sense of what quantum mechanics was actually saying. It seemed to her that physicists were fooling themselves by ignoring the foundations of their most important theory. A few years later, she had a PhD in theoretical physics but had landed a job as an assistant professor in the philosophy department at a major university, and was gaining a reputation as an expert on the Many-Worlds approach to quantum mechanics. “How do you want to do this?”

“I wrote down some questions,” he said as he pulled out his phone and pulled something up on its screen.